Twin Neighbor Embedding: A Pre-Embedding Transform for Improved Network Visualization.

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Abstract

This thesis aims to address challenges in existing methods for the visual analysis of networks by introducing the Twin Neighbor Embedding method, a simple pre-embedding transform designed to improve the quality of existing network layout algorithms. Network representations play a central role in scientific data analysis, as many real-world systems and relationships are naturally modeled as networks. Network visualization is therefore an integral component of network analysis, as spatial layouts can reveal structural properties of the underlying network in an intuitive and interpretable manner. However, existing network layout methods exhibit key limitations that can obscure structural relationships and hinder accurate interpretation of the data. We identify three recurring issues which the Twin Neighbor Embedding method aims to address: (a) vertices with small graph-theoretic distance or strong edge weights are often placed excessively close in the layout, leading to visual crowding and partial collapse of local substructure; (b) certain vertices or groups of vertices are erroneously placed close together, forming hallucinatory clusters that suggest relationships not present in the data; and (c) conventional layout methods do not explicitly encode geometric edge placement, which often results in excessive crossings of long edges. The Twin Neighbor Embedding method improves on these limitations by encoding vertex adjacency, edge adjacency, and vertex–edge incidence relationships in a unified augmented matrix used as a pre-layout transform. This representation is then supplied to standard layout algorithms in place of the traditional adjacency matrix. Despite its conceptual simplicity, the proposed transform improves layout quality across a variety of real-world and synthetic networks, as demonstrated through visual analysis and quantitative evaluation of neighborhood preservation, classification accuracy, vertex crowding and edge crossings.

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1. Introduction

1.1. Motivation

Networks have been used and studied for centuries, as they can model a variety of systems and relationships. At their most basic, networks are a set of data points (called vertices or nodes) with interconnections between them (called edges or links). The versatility of networks enables us to gain insight into systems, prove conjectures, or develop algorithms in many scientific fields.

Network analysis and scientific network visualization are an increasingly vital part of data analytics for exploratory study and confirmatory analysis of relationships, patterns, community structures, and their evolution or dynamics in real-world data networks across various domains such as biology [1], [2], [3], [4], neurology [5], sociology [6], and related domains [7], [8].

1.2. Problem Description

Network visualization conventionally is the task of assigning each vertex a position in some surface Σ. Then the edges may be drawn as lines or curves (not strictly in Σ) connecting their incident vertices. As a post-processing step, additional network properties and features may be overlaid on top of the layout as functions defined on the vertices or edges.

Denote by $G(V, E)$ the network under consideration. Each vertex $v \in V$ is mapped to a unique point $x(v) \in \Sigma$. The set $X = \{x(v) \mid v \in V\}$ is called the vertex layout or embedding[1] and is determined by a network layout algorithm. Various layout algorithms exist with differing criteria for determining the spatial configuration X. Some of these methods and criteria will be discussed in more detail in Section 2.

The surface Σ is called the embedding space and is commonly $\mathbb{R}^2$, but sometimes $\mathbb{R}^3$ or even restricted to $\mathbb{S}^1$ or $\mathbb{S}^2$ [9], [10], [11]. For the purposes of visualization Σ is low-dimensional.

[1] In topological graph theory an embedding strictly is a drawing of the vertices and edges in a way such that the edges may intersect only at their ends in the target embedding space. In this thesis embedding and layout are used interchangeably, in all cases meaning a layout.

Existing software packages for scientific network visualization differ primarily in three aspects: (i) the choice of embedding space Σ, (ii) the choice of embedding criteria and layout algorithm, and (iii) post-processing utilities and overlays. The analysis in this thesis is primarily concerned with (ii), and from this point forward it will be assumed that $\Sigma \equiv \mathbb{R}^2$ and overlays will be used only to highlight structural aspects of the network useful in the visual understanding and qualitative assessment of the layout.

Constructing such embeddings is inherently challenging. A low-dimensional layout must represent a complex network topology while preserving structural relationships that are important for visual interpretation. Because many structural constraints cannot be simultaneously satisfied in a two-dimensional embedding, layout algorithms inevitably introduce distortions whose nature depends on the chosen embedding objective.

1.3. Research Gap

Various layout algorithms exist with differing criteria for determining the spatial configuration X. Some of these methods and criteria will be discussed in more detail in Section 2.

A common criterion is to preserve the pairwise geodesic distances of vertices on G proportionally in the layout X. Yet it is impractical to rigidly impose all geodesic distance pairs be preserved in equal measures, considering the confined low-dimensional embedding space does not allow for the variety, irregularity, and uncertainty in sparse data networks. We prioritize that adjacent vertices in network G are placed in close proximity in the layout X. Neighbor preservation is the principal embedding criterion in SG-t-SNE [12], which adapts the stochastic neighbor embedding framework to graph-structured data.

However SNE-based vertex layout suffers from undesirable artifacts that obscure structural relationships and hinder accurate interpretation of the data. We identify three recurring issues: (a) vertex crowding: vertices with small graph-theoretic distance or strong edge weights are often placed excessively close in the layout, leading to visual crowding and partial collapse of local substructure ; (b) erroneous clustering: certain vertices or groups of vertices are erroneously placed close together, forming hallucinatory clusters that suggest relationships

not present in the data; and (c) excessive long-edge crossings: conventional layout methods do not explicitly encode geometric edge placement, which often results in excessive crossings of long edges.

1.4. Contributions

We introduce the Twin Neighbor Embedding (TNE) method and the TWIN software package aimed at mitigating the limitations of SNE-based methods. TNE is not a layout algorithm in the traditional sense, but instead a pre-layout transform of G which constructs a supra-network consisting of G, its line graph G_ℓ, and additional connections linking vertices of G to corresponding incident edges represented in G_ℓ. This is represented algebraically via a unified augmented matrix which encodes vertex and edge adjacency as well as vertex–edge incidence. This representation is then used in place of the traditional adjacency matrix to extend and improve pre-existing layout methods, primarily —but not limited to— SG-t-SNE. The method is described in detail in Section 3.

Despite its conceptual and mathematical simplicity, TNE proves effective in mitigating vertex crowding and erroneous clustering across various real-world and synthetic networks. Additionally, due to its explicit geometric encoding of edge placement it improves on excessive edge crossing and enables edges to be drawn as parametric curves without manual intervention by the user. Various visualization results, as well as quantitative evaluations of the method are presented in Section 4.

Our software package TWIN allows the user to interact with the Twin Neighbor Embedding method and compare to traditional vertex embedding in an interactive GUI with various tuning parameters. The package is open source and the source code can be found at https://github.com/alex-unofficial/twin-neighbor-embedding. The TWIN software package is discussed in Section 5.

Finally, we discuss the limitations of our method as well as potential future improvements in Section 6.

2. State of the Art

3. The Twin Neighbor Embedding method

4. Results and Evaluation

5. The TWIN software package and GUI

6. Conclusion and Limitations

Bibliography

[1] K. Melnyk, K. Weimann, and T. O. F. Conrad, "Understanding Microbiome Dynamics via Interpretable Graph Representation Learning," *Scientific Reports*, vol. 13, no. 1, p. 2058, Feb. 2023, doi: 10.1038/s41598-023-29098-7.

[2] H. Ashoor *et al.*, "Graph Embedding and Unsupervised Learning Predict Genomic Sub-Compartments from HiC Chromatin Interaction Data," *Nature Communications*, vol. 11, no. 1, p. 1173, Mar. 2020, doi: 10.1038/s41467-020-14974-x.

[3] F. Costa, D. Grün, and R. Backofen, "GraphDDP: A Graph-Embedding Approach to Detect Differentiation Pathways in Single-Cell-Data Using Prior Class Knowledge," *Nature Communications*, vol. 9, no. 1, p. 3685, Sept. 2018, doi: 10.1038/s41467-018-05988-7.

[4] U. Stelzl *et al.*, "A Human Protein-Protein Interaction Network: A Resource for Annotating the Proteome," *Cell*, vol. 122, no. 6, pp. 957–968, Sept. 2005, doi: 10.1016/j.cell.2005.08.029.

[5] H.-J. Park and K. Friston, "Structural and Functional Brain Networks: From Connections to Cognition," *Science*, vol. 342, no. 6158, p. 1238411, Nov. 2013, doi: 10.1126/science.1238411.

[6] C. Kadushin, Ed., *Understanding Social Networks*. New York: Oxford University Press, 2012.

[7] V. Filipov, A. Arleo, and S. Miksch, "Are We There yet? A Roadmap of Network Visualization from Surveys to Task Taxonomies," *Computer Graphics Forum*, vol. 42, no. 6, p. e14794, 2023, doi: 10.1111/cgf.14794.

[8] C. V. R. Hütter, C. Sin, F. Müller, and J. Menche, "Network Cartographs for Interpretable Visualizations," *Nature Computational Science*, vol. 2, no. 2, pp. 84–89, Feb. 2022, doi: 10.1038/s43588-022-00199-z.

[9] D. J. Watts and S. H. Strogatz, "Collective Dynamics of 'Small-World' Networks," *Nature*, vol. 393, no. 6684, pp. 440–442, 1998, doi: 10.1038/30918.

[10] J. Miller, V. Huroyan, and S. Kobourov, "Balancing Between the Local and Global Structures (LGS) in Graph Embedding," *Graph Drawing and Network*

Visualization, vol. 14465. Springer Nature Switzerland, Cham, pp. 263–279, 2023. doi: 10.1007/978-3-031-49272-3_18.

[11] Y. Lu, J. Corander, and Z. Yang, "Doubly Stochastic Neighbor Embedding on Spheres," *Pattern Recognition Letters*, vol. 128, pp. 100–106, Dec. 2019, doi: 10.1016/j.patrec.2019.08.026.

[12] N. Pitsianis, A.-S. Iliopoulos, D. Floros, and X. Sun, "Spaceland Embedding of Sparse Stochastic Graphs," in *IEEE HPEC*, 2019, pp. 1–8. doi: 10.1109/HPEC.2019.8916505.